PROJECT REPORT

AquaTrack

Water Quality Inspection & Maintenance System

Developed using Zoho Creator

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Project Title	AquaTrack – Water Quality Inspection & Maintenance System
Platform	Zoho Creator (low-code application platform)
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1. Abstract

Safe drinking water needs regular testing, inspection, cleaning and repair. In many places these are noted on paper, so records get lost and problems are not followed up. **AquaTrack** is a Zoho Creator application that keeps all these records in one place using 11 linked forms. This report explains the forms, how they connect, and clearly separates what is **implemented** (the forms) from what is only **proposed** (automation and dashboards).

Every record (test, inspection, sample, complaint and so on) is linked to a water source or equipment item, so its full history can be seen easily.

2. Introduction

2.1 Background

Water in a campus goes through borewells, tanks, pumps and filters. If any stage is not checked regularly, quality drops. Quality is judged using limits such as IS 10500:2012 for pH, TDS, turbidity and chlorine.

2.2 Problem Statement

Paper records are easy to lose and hard to search. A failed test may not be linked to the fix that followed, and cleaning dates get forgotten.

2.3 Objectives

Objectives: (1) keep a register of water sources and equipment; (2) record tests, inspections, samples and lab reports in a standard format; (3) record maintenance, cleaning and filter changes; (4) register complaints and corrective actions; (5) link every record to its source.

2.4 Scope and Status Convention

Implemented = the 11 forms and their lookup links (delivered scope). **Proposed** = automation, notifications, dashboards and reports (not claimed as completed). Sample records in this report are only examples.

3. Existing and Proposed Systems

3.1 Existing System

Readings, inspection notes and repairs are written in separate registers or spreadsheets. Lab reports are filed apart from the samples. Finding the history of one tank means searching many documents.

3.2 Proposed System

AquaTrack keeps all 11 record types in one Zoho Creator application. Each record is linked to its source or equipment. Automation and dashboards are only proposed.

Table 1: Comparison of the existing and proposed systems

Aspect	Existing (manual) system	Proposed AquaTrack system
Record storage	Paper registers, separate spreadsheets	Single Zoho Creator application (**Implemented**)
Data format	Free-form, varies by person	Fixed fields and choice lists in each form (**Implemented**)
Traceability	Manual cross-checking of documents	Lookup links to source and equipment (**Implemented**)
Follow-up of failures	Depends on memory or notes	Corrective Action form linked to the issue (**Implemented**)
Reminders and alerts	None	Automated reminders and notifications (***Proposed***)
Summary and trends	Compiled by hand	Dashboards and reports (***Proposed***)

4. Requirements and Technologies

4.1 Functional and Non-Functional Requirements

The system must let users register sources and equipment, record tests, inspections, samples, lab reports, maintenance, cleaning, filter changes and complaints, and track corrective actions. It should be easy to use in a web browser and consistent in its data. Intended users: administrator, inspectors, lab staff and maintenance staff.

Table 2: Technologies and reference standards

Component	Description and role in the project
Zoho Creator	Low-code cloud platform used to build the forms, field types, lookup relationships and record views.
Form fields	Auto-number IDs, date, number, drop-down, radio, text and lookup fields are used to model each record.
Deluge scripting	Zoho Creator's scripting language. Intended for the proposed workflow automation (***Proposed***).
Client device	Any computer or smartphone with an internet connection and a modern web browser.
Reference standards	BIS IS 10500:2012 and WHO Guidelines for Drinking-water Quality, used as reference limits when testers judge results.

5. System Architecture

AquaTrack has three layers provided by Zoho Creator: the **web interface** where users enter data, the **forms and relationships** that define the data, and the **cloud database** that stores it. No separate server is needed.

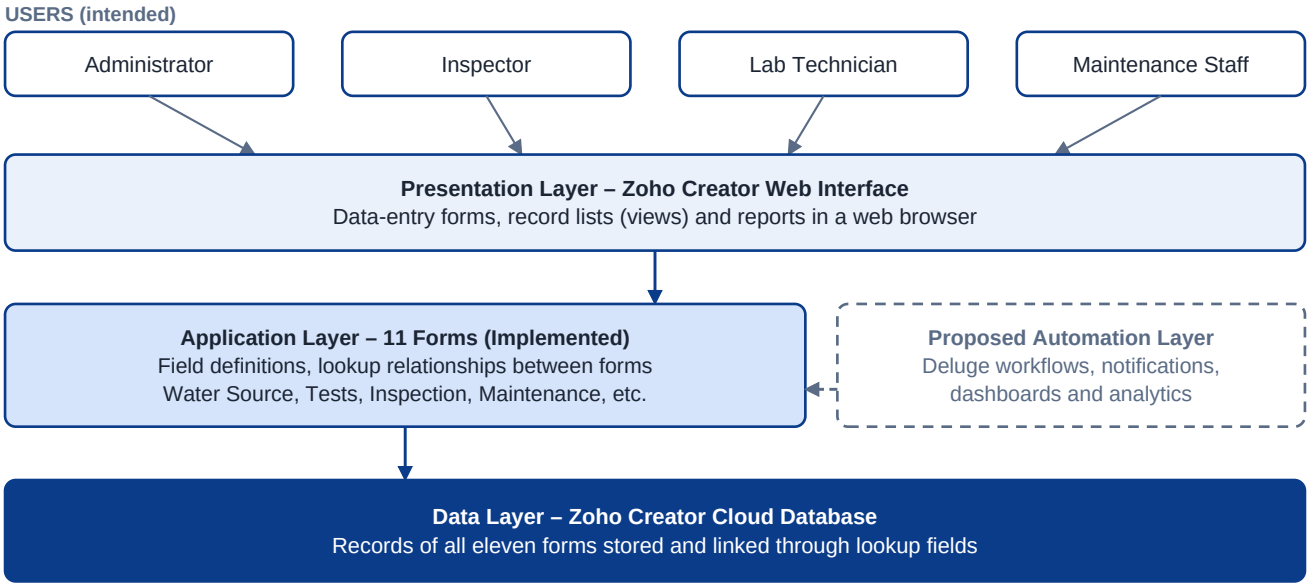


Figure 1: Layered architecture of AquaTrack (solid = implemented, dashed = proposed)

The dashed box is not built yet. It would use Deluge scripts to send alerts, for example when a test fails.

5.1 Data Flow Summary

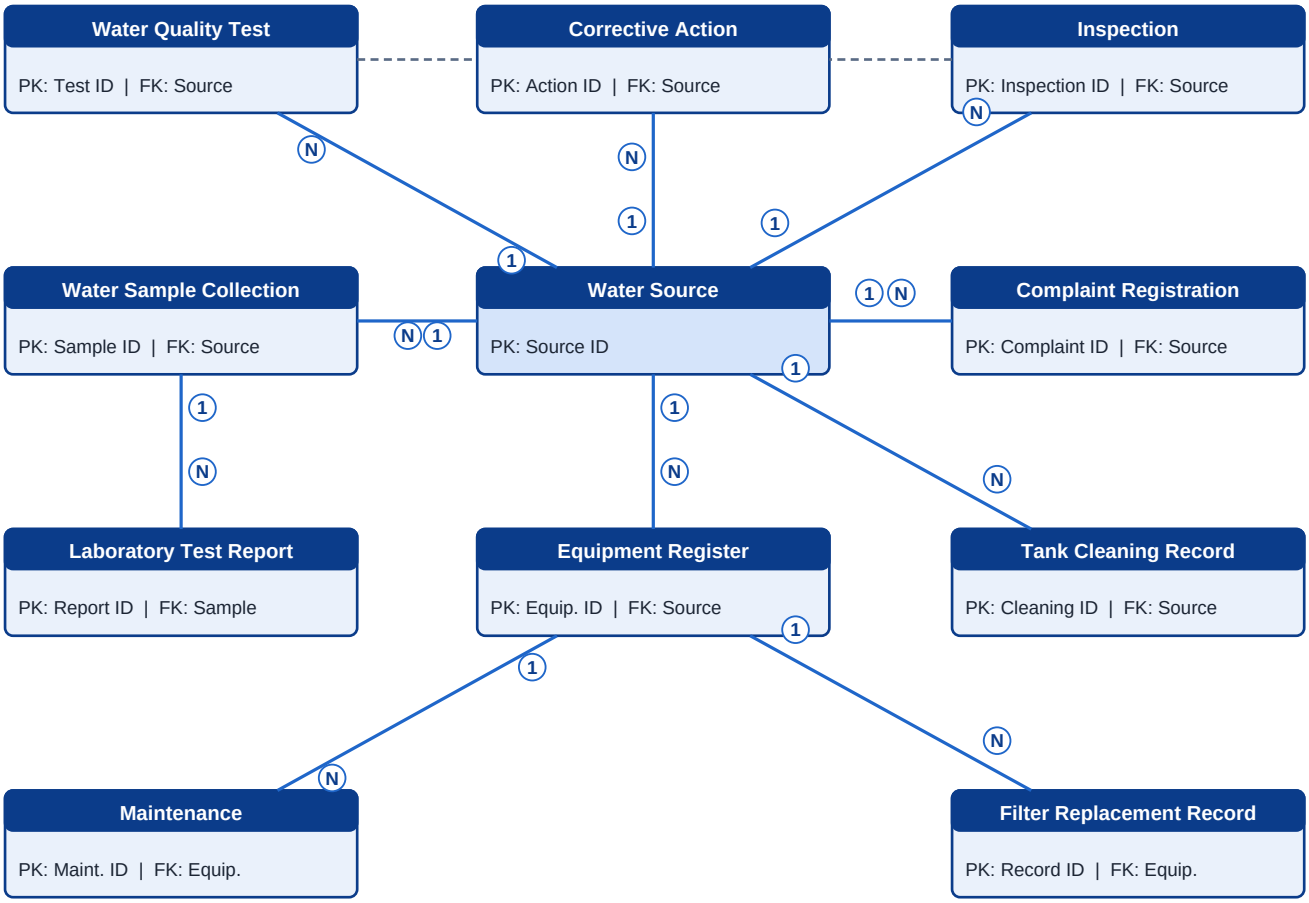
First a user registers a source or equipment. Every later record is created by choosing that source from a drop-down list. Figure 2 shows how the forms are related.

Table 3: Layers, components and status

Layer	Component	Responsibility	Status
Presentation	Zoho Creator web forms and record lists	Data entry and viewing of stored records	**Implemented**
Application	Eleven forms with lookup fields	Define fields, choice lists and relationships	**Implemented**
Data	Zoho Creator cloud database	Store all submitted records	**Implemented**
Automation	Deluge workflows and notifications	Auto-create actions, send reminders	***Proposed***
Reporting	Dashboards and reports	Summaries, trends and due lists	***Proposed***

6. ER Diagram

Water Source is the centre: tests, inspections, samples, complaints, tank cleaning, equipment and corrective actions all link to it (one source, many records). **Equipment Register** links to Maintenance and Filter Replacement, and **Water Sample Collection** links to Laboratory Test Report.



Solid line: mandatory lookup (1 : N)
Dashed line: optional reference from Corrective Action

Figure 2: Entity-relationship diagram of the eleven AquaTrack forms

Table 4: Relationships implemented through lookup fields

Parent form	Child form	Cardinality	Meaning
Water Source	Test, Inspection, Sample, Complaint, Tank Cleaning, Equipment, Corrective Action	1 : N	A source can have many records of each type.
Water Sample Collection	Laboratory Test Report	1 : N	A laboratory report refers to the sample it analysed.
Equipment Register	Maintenance, Filter Replacement Record	1 : N	Servicing history is kept per equipment item.
Water Quality Test / Inspection	Corrective Action (optional)	1 : N	An action may cite the test or inspection that triggered it.

7. Application Forms (Implemented)

Each form below has its purpose, main fields and a sample entry (sample values are illustrative).

7.1 Water Source Form

Master list of every borewell, sump, tank or supply point. All other forms refer to it.

Principal fields: Source ID, Name, Type, Location, Capacity (L), Commissioned Date, Status.

Example: *WS-001, Main Overhead Tank – Block A, 10,000 L, Active.*

7.2 Water Quality Test Form

Records on-site readings for a source. Reference limits (IS 10500:2012): pH 6.5–8.5, turbidity 1 NTU (5 permissible), chlorine at least 0.2 mg/L.

Principal fields: Test ID, Water Source (lookup), Date, pH, TDS, Turbidity, Residual Chlorine, Result, Tested By.

Example: *WQT-001, WS-001, 02-Sep-2026, pH 7.2, TDS 310, Turbidity 0.8, Pass.*

7.3 Inspection Form

Records the physical condition of a source: lid, leakage, pipes and cleanliness.

Principal fields: Inspection ID, Water Source (lookup), Date, Inspector, Leakage, Cover Condition, Remarks, Overall Condition.

Example: *INS-001, WS-001, 16-Sep-2026, tank lid loose, Needs Attention.*

7.4 Maintenance Form

Logs preventive and breakdown work on equipment such as pumps and chlorinators.

Principal fields: Maintenance ID, Equipment (lookup), Type, Scheduled Date, Completed Date, Technician, Description, Status.

Example: *MNT-001, EQ-001 Booster Pump, Preventive, Completed.*

7.5 Corrective Action Form

Documents the fix after a failed test, inspection finding or complaint, until it is closed.

Principal fields: Action ID, Water Source (lookup), optional Test/Inspection lookups, Issue, Action Taken, Assigned To, Due Date, Closure Date, Status.

Example: *CA-001, WS-001, drain and clean tank, due 20-Sep-2026, Closed.*

7.6 Water Sample Collection Form

Records when, where and by whom a sample was collected for the lab.

Principal fields: Sample ID, Water Source (lookup), Date/Time, Collected By, Sample Point, Container, Status.

Example: *SMP-001, WS-001, 15-Sep-2026, outlet tap Block A, Sent to Lab.*

7.7 Laboratory Test Report Form

Stores the lab's results, linked to the sample it analysed.

Principal fields: Report ID, Sample (lookup), Lab Name, Report Date, pH, TDS, Turbidity, Hardness, Coliform, Conclusion.

Example: *LAB-001, SMP-001, Turbidity 6.5 NTU, Does not conform.*

7.8 Tank Cleaning Record Form

Records tank cleaning date, method, disinfectant and the next due date (reminders are only proposed).

Principal fields: Cleaning ID, Water Source (lookup), Date, Method, Disinfectant, Cleaned By, Next Due Date.

Example: *TC-001, WS-001, 20-Sep-2026, next due 20-Mar-2027.*

7.9 Filter Replacement Record Form

Tracks replacement of sediment, carbon and RO filters on equipment.

Principal fields: Record ID, Equipment (lookup), Filter Type, Date, Old Filter Condition, Replaced By, Next Date.

Example: *FR-001, EQ-002 RO Plant, Sediment filter, 05-Oct-2026.*

7.10 Complaint Registration Form

Registers a water complaint with its type, priority and status.

Principal fields: Complaint ID, Name, Contact, Water Source (lookup), Type, Description, Date, Priority, Status.

Example: *CMP-001, 14-Sep-2026, Block A, Turbidity, High, Open.*

7.11 Equipment Register Form

Second master list: pumps, RO plants, chlorinators and filter units with serial number and warranty.

Principal fields: Equipment ID, Name, Type, Make/Model, Serial No., Installation Date, Water Source (lookup), Warranty, Status.

Example: *EQ-002, RO Plant, WS-002 (Borewell – Hostel), Active.*

Table 5: Summary of form relationships to master data

Form	Key lookup	Form	Key lookup
Water Quality Test	Water Source	Tank Cleaning Record	Water Source
Inspection	Water Source	Filter Replacement Record	Equipment Register
Maintenance	Equipment Register	Complaint Registration	Water Source
Corrective Action	Water Source (+ optional)	Laboratory Test Report	Water Sample Collection
Water Sample Collection	Water Source	Equipment Register	Water Source

8. Workflow Design

Problems found by a failed test, bad lab report, complaint or inspection lead to a Corrective Action. The action is done through maintenance, cleaning or filter change, and closed after a re-test.

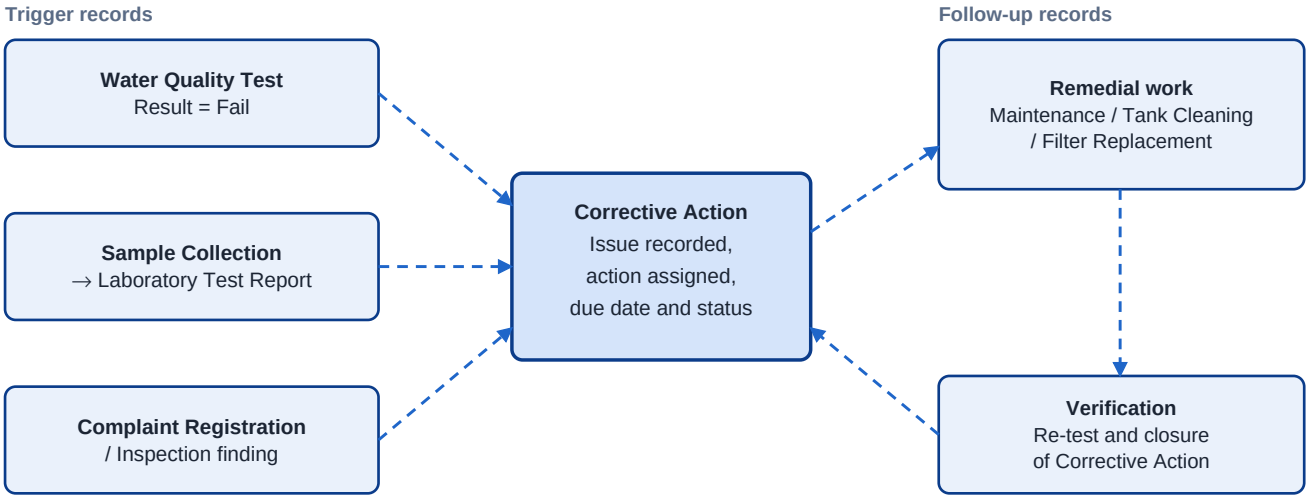


Figure 3: Intended corrective workflow (dashed arrows show the process; automatic triggering is proposed)

Table 6: Status of workflow elements

Workflow element	Description	Status
Linking records	Users select the related source, equipment or sample through lookup fields in each form.	**Implemented**
Manual follow-up	Users create the Corrective Action and remedial records themselves after a problem is found.	**Implemented**
Auto-creation of actions	A Corrective Action would be created automatically when a test result is Fail.	***Proposed***
Due-date reminders	Alerts for tank cleaning, filter replacement and open actions.	***Proposed***
Approval and escalation	Escalation of overdue actions to the administrator.	***Proposed***

9. Dashboard and Reports

The stored data can be turned into reports for supervisors. The reports below are **proposed** and not yet completed.

Table 7: Proposed reports and dashboard components

Report / component	Source form(s)	Purpose	Status
Source register summary	Water Source	List of sources by type, capacity and status	***Proposed***
Test results by source	Water Quality Test	Parameter values over time for one source	***Proposed***
Failed test report	Water Quality Test	Records where Result = Fail	***Proposed***
Open corrective actions	Corrective Action	Actions not yet closed, ordered by due date	***Proposed***
Maintenance due list	Tank Cleaning, Filter Replacement	Records whose next due date is near	***Proposed***
Complaint status summary	Complaint Registration	Counts by type, priority and status	***Proposed***

10. Implementation

The two master forms (Water Source and Equipment Register) come first because other forms depend on them. Each form uses an auto-number ID, date fields, number fields for readings, drop-downs for status and type, and lookup fields to the master forms. Workflow scripts, notifications and dashboards are not implemented yet.

11. Testing

Testing checks form behaviour and links. The table lists each test with its expected result; the actual outcome must be confirmed by running it in the application.

Table 8: Functional test cases for the implemented forms

ID	Scenario	Input	Expected result
T1	Register a source	Fill Water Source form with sample data	Record saved with an auto-number Source ID
T2	Lookup integrity	Open Water Quality Test and click Water Source	Only registered sources are listed
T3	Numeric entry	Enter pH 7.2, TDS 310, Turbidity 0.8	Values are stored as numbers
T4	Required field	Submit Complaint without a description	Form prompts for the missing field
T5	Sample-to-report link	Create Laboratory Test Report selecting SMP-001	Report is linked to the sample
T6	Equipment link	Create Filter Replacement for EQ-002	Record shows the equipment selected
T7	Status update	Edit Corrective Action: Open to Closed, add closure date	Updated values are saved

12. Results

The table follows one sample incident, muddy water at Block A, through the forms. All values are illustrative sample data.

Table 9: Illustrative end-to-end trace using sample records

Step	Form	Record	Key data
1	Water Quality Test	WQT-001	02-Sep-2026, WS-001, Turbidity 0.8 NTU, Pass (baseline)
2	Complaint Registration	CMP-001	14-Sep-2026, WS-001, muddy water, High priority
3	Water Sample Collection	SMP-001	15-Sep-2026, sample sent to laboratory
4	Inspection	INS-001	16-Sep-2026, tank lid loose, Needs Attention
5	Laboratory Test Report	LAB-001	17-Sep-2026, Turbidity 6.5 NTU, Does not conform
6	Corrective Action	CA-001	Drain and clean tank, due 20-Sep-2026
7	Tank Cleaning Record	TC-001	20-Sep-2026, next due 20-Mar-2027
8	Water Quality Test	WQT-002	22-Sep-2026, Turbidity 0.8 NTU, Pass; CA-001 closed

Result: every record points to the same source, giving a clear history. Alerts and dashboards are not part of these results.

13. Conclusion

AquaTrack shows that water quality inspection and maintenance records can be organised on a low-code platform. The 11 forms and their links give a traceable history of each source and equipment item. The delivered scope is the forms and relationships; automation, notifications and dashboards are proposed work.

13.1 Limitations

Users must pick the correct source and create follow-up records themselves. There are no reminders or dashboards yet, and sample records here are illustrative.

14. Future Enhancements (Proposed)

These enhancements are proposed only and are not implemented.

Enhancement	Description
Automatic corrective actions	Create a Corrective Action record automatically when a test or laboratory result fails.
Due-date reminders	Email or mobile alerts for tank cleaning, filter replacement and overdue actions.
Dashboard and reports	Build the dashboard and reports listed in Table 7, with trend charts for pH, TDS and turbidity.
Role-based access	Restrict data entry and approval rights separately for inspectors, laboratory staff and administrators.
Mobile field entry	Use the platform's mobile access for on-site entry, with photographs of inspection findings.
Asset identification	QR-code labels on tanks and equipment so that a record can be opened by scanning.
Sensor integration	Long-term option to import readings from online water quality sensors.

15. References

[1] Bureau of Indian Standards, *IS 10500:2012 – Drinking Water Specification*, New Delhi, 2012.

[2] World Health Organization, *Guidelines for Drinking-water Quality: Fourth Edition Incorporating the First and Second Addenda*, Geneva, 2022.

[3] Zoho Corporation, *Zoho Creator Help Documentation*, https://www.zoho.com/creator/help/.

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[5] Central Public Health and Environmental Engineering Organisation (CPHEEO), *Manual on Water Supply and Treatment*, Ministry of Urban Development, Government of India, 1999.